



Semester II 2023/2024

Subject : PROBABILITY & STATISTICAL DATA ANALYSIS
(SECI / SCSi 2143/ SECI 1143)
Task : Chapter 3 & Chapter 4
Due Date : **Week 9 (12 May 2024)**

INSTRUCTION:

1. This is a **GROUP** assignment. Please clearly write the group members **NAME & MATRIC NUMBER** in the front page of the submission.
 2. This assignment contributes to 5% of overall course marks.
 3. Only **HANDWRITTEN** submission is accepted:
 - a. Submissions using any reporting or statistical tools (e.g.: MS Word, MS Excel, etc.,) will be **REJECTED**.
 - b. Make sure the submission is neatly written. Any submission with handwriting that is unreadable, will be **REJECTED**.
 - c. For answer that need to draw graphs, using graph paper is optional. You can use plain paper.
 - d. Round your answers to **THREE** decimal places.
 - e. Please scan/snapshot your work and save as a PDF file.
 4. Submission via eLearning – only **ONE** group member needs to submit on behalf of the group.
-

QUESTION 1[12 MARKS]

Table 1 shows data recorded on height in cm of 60 students from Sekolah Menengah Taman Bukit Dahlia.

Table 1

Class Interval	Frequency
$150 \leq x < 160$	12
$160 \leq x < 170$	20
$170 \leq x < 180$	5
$180 \leq x < 190$	3

Class Interval	Class midpoint	Frequency	cu. Frequency.
$150 \leq x < 160$	155	12	12
$160 \leq x < 170$	165	20	32
$170 \leq x < 180$	175	5	37
$180 \leq x < 190$	185	3	40

$$\begin{aligned}
 \text{a) mean} &= \frac{155(12) + 165(20) + 175(5) + 185(3)}{40} \\
 &= \frac{6590}{40} \\
 &= 164.75
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } i &= \frac{40}{2} \\
 &= 20^{\text{th}}.
 \end{aligned}$$

median class : $160 \leq x < 170$

$$\begin{aligned}
 \text{c) mode} &= l + h \times \frac{(f_1 - f_0)}{(2f_1 - f_0 - f_2)} \\
 &= 160 + 10 \times \frac{(20 - 12)}{(40 - 12 - 5)} \\
 \text{mode} &= 162.61
 \end{aligned}$$

d) modal class : $160 \leq x < 170$

Based on the above value, calculate the:

- a) Mean (4 Marks)
- b) Median (4 Marks)
- c) Mode (3 Marks)
- d) Modal class (1 Mark)

QUESTION 2 [15 MARKS]

Imagine that you are a teacher collecting data on the scores of 10 students in a recent mathematics quiz. The scores obtained are as follows: 85, 90, 75, 88, 92, 80, 85, 82, 90, and 85. Your goal is to analyze the central tendencies of the quiz scores.

- a) Calculate the mean, median, and mode of the quiz scores. (5 Marks)
- b) Interpret what these values (*mean, median, and mode*) suggest about the performance of the students in the quiz. Based on this value which statistic that more appropriate represent the summary of the score. (4 Marks)
- c) Later you rely that some of the scores are wrongly recorded and new scores are as follows:
55, 65, 65, 70, 85, 95, 95, 95, 100, 100
 - i. Calculate new mean, median and mode for this dataset.
 - ii. Plot the graph of central tendency.
 - iii. Evaluate the exact performance of the student of this data compared to previous data.

(6 Marks)

2a) 85 90 75 88 92 80 85 82 90 85

$$\begin{aligned}\text{mean} &= \frac{85 + 90 + 75 + 88 + 92 + 80 + 85 + 82 + 90 + 85}{10} \\ &= \frac{852}{10} \\ &= 85.2.\end{aligned}$$

75 80 82 85 85 85 88 90 90 92

$$\begin{aligned}\text{median} &= \frac{85 + 85}{2} \\ &= 85.\end{aligned}$$

$$\text{mode} = 85.$$

b) the median could know the average scores the students get.

Teacher could easily know the standard and understanding of students in the course.

This is the best statistic to know the average of the scores.

The median statistic could only know the middle score that got by student in the class.

It didn't help much in summarising the score.

While for the mode, teachers could know score mostly get by the students only.

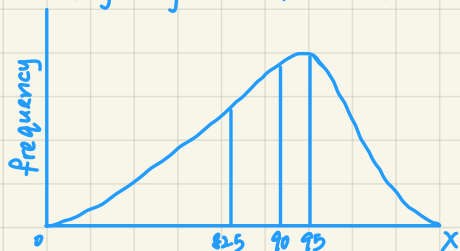
c) 55 65 65 70 85 95 95 95 100 100

$$\begin{aligned}\text{i) mean} &= \frac{55 + 65 + 65 + 70 + 85 + 95 + 95 + 95 + 100 + 100}{10} \\ &= \frac{825}{10} \\ &= 82.5\end{aligned}$$

$$\begin{aligned}\text{Median} &= \frac{85 + 95}{2} \\ &= 90\end{aligned}$$

$$\text{mode} = 95$$

ii) Negatively Skewed Distribution



iii) Students in previous data get a better average score in the class.

The average score in the previous data is better than the second data.

The dispersion of data in second data is bigger which shows previous data is more average and consistent.

The range of second data is bigger compare to the previous data.

QUESTION 3 [13 MARKS]

Suppose you are analyzing the monthly sales data for a small business over the past year. You have collected the total sales figures for each month from January to December. Table 2 shows the sales figures (in thousands of Ringgit Malaysia) are as follows:

Table 2

Months	Sales Figure	Months	Sales Figure
January	25	July	40
February	30	August	38
March	28	September	33
April	35	October	36
May	32	November	31
June	27	December	29

$$\mu = \frac{25 + 30 + 28 + 35 + 32 + 27 + 40 + 38 + 33 + 36 + 31 + 29}{12}$$

a) Calculate the monthly sales figures for:

$$= 32$$

i. Range

$$40 - 25 = 15$$

(2 Marks)

ii. Variance

$$s^2 = \frac{\sum (x_i - \mu)^2}{N} = \frac{(25-32)^2 + (30-32)^2 + (28-32)^2 + (35-32)^2 + (32-32)^2 + (27-32)^2 + (40-32)^2 + (38-32)^2 + (33-32)^2 + (36-32)^2 + (31-32)^2 + (29-32)^2}{12}$$

(5 Marks)

iii. Standard deviation

$$= 19.167$$

$$\sqrt{19.167} = 4.378$$

(1 Mark)

b) Interpret the above calculation to understand the variability in the business's monthly sales performance throughout the year.

(3 Marks)

c) Discuss how this understanding can inform strategic decisions for managing the business's finances and operations.

(2 Marks)

QUESTION 4 [10 MARKS]

A company wants to assess the level of effectiveness of a new training program designed to improve employees' productivity. The productivity scores are normally distributed with a mean of 50 units and a standard deviation of 10 units.

- What percentage of employees in the sample showed an increase in productivity after the training program?
- What is the probability of employees having a productivity score between 37 and 65 after the training?
- Suppose that the company has 1000 number of employees and plan to conduct a team building for those who not achieved minimum of score of 20 units. The cost for team building is RM200 per head. How much the company need to estimate the budget for the team building cost.
- If the company wants to identify the top 5% of employees with the highest productivity scores after the training, what minimum score would an employee need to achieve?

QUESTION 5 [15 MARKS]

In the Information Technology course, students take an exam with 6 multiple-choice questions. Each question has 4 options, with only one correct answer. The number of correct answers a student gets on the exam represents their performance in the course.

$p = 0.25$ $q = 0.75$

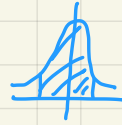
- Identify the random variable X .
- Construct up the table for the probability of X .
- What is the mean of this distribution?
- Determine the probability of a student getting at least 3 correct answers on the exam.
- What is the probability that the first incorrect answer occurs in the fourth question?

~~e)~~ $\rightarrow$ geometric

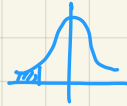
$$4) a) P(X > 50) = P\left(Z > \frac{50-50}{10}\right) \\ = P(Z = 0) \\ = 0.500$$

$$\text{percentage} = 0.5 \times 100\% \\ = 50\%$$

$$b) P(37 < X < 65) = P\left(\frac{37-50}{10} < Z < \frac{65-50}{10}\right) \\ = P(-1.30 < Z < 1.50)$$



$$P(Z > -1.30) = 0.5 - 0.4032 \\ = 0.0968$$



$$P(Z < 1.50) = 0.5 + 0.4332 \\ = 0.9332$$



$$P(-1.30 < Z < 1.50) = 0.9332 - 0.0968 \\ = 0.8364$$

$$c) P(X < 20) = P\left(Z < \frac{20-50}{10}\right) \\ = P(Z < -3) \\ = 0.5 - 0.4487 \\ = 0.0013$$



$$\text{tot people} = 0.0013 (1000) \\ = 1.3 \\ \approx 2$$

$$\text{tot budget} = 2 \times 200 \\ = \text{RM } 400$$

$$d) P(X > a) = 0.05$$

$$0.5 - P(X < a) = 0.05$$

$$P(X < a) = 0.45$$

$$\frac{a - \mu}{\sigma} = 1.64$$

$$a = 1.64(10) + 50$$

$$a = 66.4$$

$$X = 66.4$$



5a) X is the number of correct ans get.

b) X	0	1	2	3	4	5	6
	${}^6C_0 (0.25)^0$	${}^6C_1 (0.25)^1$	${}^6C_2 (0.25)^2$	${}^6C_3 (0.25)^3$	${}^6C_4 (0.25)^4$	${}^6C_5 (0.25)^5$	${}^6C_6 (0.25)^6$
$P(X)$	$(0.75)^6$	$(0.75)^5$	$(0.75)^4$	$(0.75)^3$	$(0.75)^2$	$(0.75)^1$	$(0.75)^0$
	$= 0.178$	$= 0.356$	$= 0.297$	$= 0.132$	$= 0.033$	$= 0.004$	$= 2.44 \times 10^{-4}$

c) $\mu = np$
 $= 6 (0.25)$
 $= 1.500$

d) $P(X \geq 3) = 0.132 + 0.033 + 0.004 + 2.44 \times 10^{-4}$
 $= 0.169$

~~e) $P(X = 4) = (1 - 0.75)^{4-1} (0.75)$~~
 ~~$= 0.012$~~

QUESTION 6 [10 MARKS]

In a coffee shop, customers arrive at random intervals throughout the day. The shop owner is interested in understanding the arrival patterns to better manage staffing levels. The probability of a customer ordering a cappuccino is 0.70. X represents the number of customers that will arrive until the 4th customer who orders a cappuccino.

- a) What is the probability distribution of X ?
- b) What is the standard deviation of this distribution?
- c) Determine the probability that exactly 6 customers will arrive until the 4th customer who orders a cappuccino.
-  d) Determine the probability that exactly 7 cappuccinos will be ordered within the next 12 customer arrivals.

$$\begin{aligned} P(X=7) &= {}^4C_7 (0.7)^7 (0.3)^5 \\ &= 0.138 \end{aligned}$$

$$6a) P(X = k) = \binom{k-1}{4-1} \times (0.7)^4 (1-0.7)^{k-4}$$

This represents the probability of having k customers before the 4th customer who orders a cappuccino

$$\begin{aligned} b) \sigma_x &= \sqrt{\frac{(1-p)}{p^2}} \\ &= \sqrt{\frac{(1-0.7)}{(0.7)^2}} \\ &= 0.783 \end{aligned}$$

$$\begin{aligned} c) b^*(6, 4, 0.7) &= \binom{6-1}{4-1} (0.7)^4 (0.3)^{6-4} \\ &= {}^5C_3 (0.7)^4 (0.3)^2 \\ &= 0.216 \end{aligned}$$

d)